Lab # 1

Introduction to Linux operation system

I. THEORY

This is an introduction to the major features of Linux that helps to get acquainted with them. It does not go into detail or cover any advanced topics, as is done in later lab exercises. Instead it is intended to give a head start in understanding what Linux is, what Linux offers, and what is needed to run it. Linux is the clone of UNIX operating system that runs on machines with an Intel 80386 processor or better, as well as Intel-compatible CPU.

Linux is not UNIX, as UNIX is a copyrighted piece of software that demands license fees when any part of its source code is used. Linux was written from scratch to avoid license fees entirely though the operation of the Linux operating system is based entirely on UNIX. It shares UNIX's command set and look-and-feel, so if you know either UNIX or Linux, you know the other too.

Linux supports a wide range of software, from TeX(a text formatting language) to (a graphical user interface) to the GNU C/C++ compilers to TCP/IP networking.

II. Major Features of Linux

- Multi-user: Multiple users can run Linux side by side.
- Multitasking: Run spreadsheet, Email, browse internet on the same time.
- Wide variety of compatible software.
- Ability to run on older hardware save upgrade cost.
- Easy to learn, similar features to Windows.

III. Major Distributions

- Top Ten Distributions are:

1. Ubuntu
2. Fedora
3. Open SUSE
4. Debian
5. Mandriva
6. Slackware
7. Gentoo
8. CentOS

- In the Enterprise

1. Red Hat Enterprise Linux
2. Novell SUSE Linux Enterprise

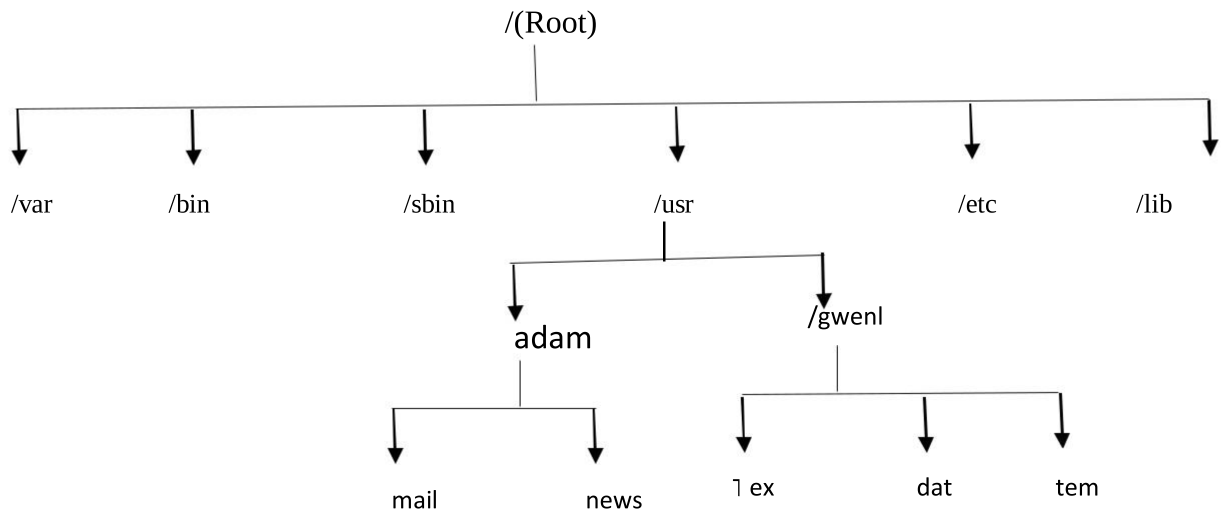
IV. What comes with Linux OS

- Every Linux distributions comes with basic required softwares like Photo Viewer, Video player, Document Editor, Text Editor, Web Browser, Games etc.
- Many Linux distributions come with programming languages installed like PHP, Java, C, C++ etc.

V. Linux File System

Under Linux, the file space that is visible to users is based on an inverted tree structure, with the root at the top. The various directories and files in this space branch downwards from the root. The top directory ‘/’ is known as the Root directory.

The Linux operating system is made up of several directories and many different files. Depending on how you selected your installation, these directories have different file systems. Typically most of the operating systems resides on two file systems: the root file system known as ‘/’ and a file system mounted under/user.



1. / – Root

- Every single file and directory starts from the root directory.
- Only root user has write privilege under this directory.

2. /bin – User Binaries

- Contains binary executables.
- Common linux commands you need to use in single-user modes are located under this directory.
- Commands used by all the users of the system are located here. For example: ps, ls, ping, grep, cp.

3. /sbin – System Binaries

Just like /bin, /sbin also contains binary executables.
But, the linux commands located under this directory are used typically by system administrator, for system maintenance purpose. For example: iptables, reboot, fdisk, ifconfig, swapon

4. . /etc – Configuration Files

- Contains configuration files required by all programs.
- This also contains startup and shutdown shell scripts used to start/stop individual programs.
- For example: /etc/resolv.conf, /etc/logrotate.conf

5. . /dev – Device Files

- Contains device files.
- These include terminal devices, usb, or any device attached to the system.
- For example: /dev/tty1, /dev/usbmon0

6. . /proc – Process Information

- Contains information about system process.
- This is a pseudo filesystem contains information about running process.
- For example: /proc/{pid} directory contains information about the process with that particular pid.
- This is a virtual filesystem with text information about system resources. For example: /proc/uptime

7. . /var – Variable Files

- var stands for variable files.
- Content of the files that are expected to grow can be found under this directory.
- This includes — system log files (/var/log); packages and database files (/var/lib); emails (/var/mail); print queues (/var/spool); lock files (/var/lock); temp files needed across reboots (/var/tmp);

8. /tmp – Temporary Files

- Directory that contains temporary files created by system and users.
- Files under this directory are deleted when system is rebooted.

9. . /usr – User Programs

- Contains binaries, libraries, documentation, and source-code for second level programs.
- /usr/bin contains binary files for user programs.
- /usr/sbin contains binary files for system administrators
- /usr/local contains users programs that you install from source. For example, when you install apache from source, it goes under /usr/local/apache2

10. /home – Home Directories

- Home directories for all users to store their personal files.
- For example: /home/john, /home/nikita

11. /boot – Boot Loader Files

Contains boot loader related files.

Some of the frequently used Linux commands are : su , pwd , cd , ls , more , less , find , grep and man.

VI. Executing a Linux commands

1. pwd

Use the pwd command to print the working directory (the current directory you are in).

Example

$ pwd
/home/john

2. cd

You can change your current directory with the cd command (Change Directory).

Example

$ cd /etc
$ pwd /etc

3. cd ..

To go to the parent directory (the one just above your current directory in the directory tree), type cd .. .

Example

cd ..
$ pwd /usr/share/games
$ cd ..
$ pwd /usr/share

4. cd –

Another useful shortcut with cd is to just type cd - to go to the previous directory.

Example

$ pwd
/home/bilal
$ cd /etc
$ pwd
/etc
$ cd – /home/bilal
$ cd –
/etc

5. ls

You can list the contents of a directory with ls.

Example

$ ls
allfiles.txt dmesg.txt httpd.conf stuff summer.txt

6. ls –a

A frequently used option with ls is -a to show all files. Showing all files means including the hidden files.

Example

- $ ls
  allfiles.txt dmesg.txt httpd.conf stuff summer.txt
- $ ls -a
  allfiles.txt .bash_profile dmesg.txt .lesshst stuff .bash_history .bashrc httpd.conf .ssh summer.txt

7. ls –l

Many times you will be using options with ls to display the contents of the directory in different formats or to display different parts of the directory. Typing just ls gives you a list of files in the directory. Typing ls -l (that is a letter L, not the number 1) gives you a long listing.

Example

$ ls -l total
23992
-rw-r--r-- 1 bilal bilal 24506857 2006-03-30 22:53 allfiles.txt
-rw-r--r-- 1 bilal bilal 14744 2006-09-27 11:45 dmesg.txt -rw-r--r-
- 1 bilal bilal 8189 2006-03-31 14:01 httpd.conf drwxr-xr-x 2
bilal bilal 4096 2007-01-08 12:22 stuff -rw-r--r-- 1 bilal bilal
0 2006-03-30 22:45 summer.txt

8. ls –lh

Another frequently used ls option is -h. It shows the numbers (file sizes) in a more human readable format.

Eample

$ ls -lh total
24M
-rw-r--r-- 1 bilal bilal 24M 2006-03-30 22:53 allfiles.txt
-rw-r--r-- 1 bilal bilal 15K 2006-09-27 11:45 dmesg.txt
-rw-r--r-- 1 bilal bilal 8.0K 2006-03-31 14:01 httpd.conf drwxr-
xr-x 2 bilal bilal 4.0K 2007-01-08 12:22 stuff -rw-r--r-- 1 bilal
bilal 0 2006-03-30 22:45 summer.txt

VII. Creating a Directory

1. mkdir -p

To create a new directory, use the mkdir command. The syntax is mkdir <name>, where <name> is replaced by whatever you want the directory to be called. This creates a subdirectory with the specified name in your current directory:

Example

$ mkdir MyDir

$ cd MyDir

$ls

2. mkdir -p

When given the option -p, then mkdir will create parent directories as needed.

Example

$ mkdir -p MyDir2/MySubdir2/ThreeDeep

ls MyDir2

MySubdir2

$ ls MySubdir2

ThreeDeep

VIII. Removing Directories

1. rmdir

When a directory is empty, you can use rmdir to remove the directory. Before you can remove a directory, it must be empty (the directory can't hold any files or subdirectories).

Otherwise, you see rmdir: <directory>:

Directory not empty

2. rmdir –p

And similar to the mkdir -p option, you can also use rmdir to recursively remove directories.

Eample

$ mkdir -p dir/subdir/subdir2

$ rmdir -p dir/subdir/subdir2

Tasks

1. List all files in the current directory.

2. List all files in human readable form.

3. Display your current directory.

4. Change to the / directory.

5. Go to the ~ directory from the root directory.

6. List the contents of the root directory.

7. List the files in /boot in a human readable format.

8. Create a directory testdir in your ~ directory.

9. Create in one command the directories ~/dir1/dir2/dir3 (dir3 is a subdirectory from dir2, and dir2 is a subdirectory from dir1).

10. Remove the directory testdir.

11. Move TesingDir from ~ to ~/Tests directories.

12. Rename a directory from TesingDir to testingDir